Real-Time Canadian SME Credit Risk Analytics Pipeline

Course Name: Big Data System Design

Course Code: MBAI 5110G

Term & Year: Winter 2026

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1. Problem Statement and Dataset

1.1 Problem Statement

Canadian small and medium-sized enterprises (SMEs) represent over 98% of all businesses in Canada and account for approximately 68% of total private-sector employment. Yet access to timely and accurate credit assessment remains a critical bottleneck for SME lenders. Traditional credit underwriting processes rely on slow, paper-driven workflows that cannot keep pace with the volume and velocity of modern loan applications.

This project addresses a real-world challenge faced by Canadian fintech lenders such as Borrowell and Uplinq: how to assess SME creditworthiness quickly, accurately, and at scale. The pipeline is framed within the context of Canada's emerging Open Banking framework, which mandates greater data transparency and positions data-driven credit decisioning as both a regulatory requirement and a competitive differentiator.

The key analytical objectives of this project are to identify default rate trends across industries and geographic regions; to apply window functions to detect year-over-year credit quality deterioration by loan grade; to rank and percentile-score borrowers within peer groups for targeted risk stratification; to simulate real-time loan application ingestion and risk scoring via a streaming pipeline; and to store and query processed results in a NoSQL database for downstream reporting.

1.2 Dataset

The primary dataset used in this project is the Lending Club Loan Dataset, sourced from Kaggle, a reputable open-data platform endorsed in the course guide. The dataset covers accepted loan applications from 2007 to Q4 2018 and is publicly available at [https://www.kaggle.com/datasets/wordsforthewise/lending-club.](https://www.kaggle.com/datasets/wordsforthewise/lending-club)

Attribute	Details
File Name	accepted_2007_to_2018Q4.csv
Records	Approximately 2.26 million loan applications
Time Span	January 2007 to December 2018 (Q4)
Format	CSV, loaded into Spark via simulated HDFS path
Key Features	loan_amnt, int_rate, grade, emp_length, annual_inc, dti,

	fico_range_low, delinq_2yrs, revol_util, loan_status, addr_state, purpose, issue_d
Target Variable	default (binary): 1 = Charged Off or Default; 0 = Fully Paid
Source	Kaggle, publicly available and open-licensed

Records with ambiguous loan status such as Current, Late, or In Grace Period were excluded during data cleaning to ensure a clean binary label for model training.

2. Tools and Architecture

2.1 Architecture Overview

The pipeline follows a Lambda Architecture pattern, which separates data processing into three distinct layers: Batch, Speed, and Serving. This design ensures both high-throughput historical analysis and low-latency real-time scoring are supported within a unified system.

Layer	Technology	Responsibility
Batch Layer	Apache Spark (PySpark)	Ingest CSV, clean, transform, apply window functions, train XGBoost model, store Parquet
Speed Layer	Kafka (simulated) and Spark Structured Streaming	Ingest live loan applications, apply real-time XGBoost scoring, detect model drift
Serving Layer	MongoDB (mongomock)	Store batch aggregates and real-time scores for unified querying and reporting

2.2 Tool Selection and Justification

Apache Spark (PySpark)

Apache Spark serves as the core distributed processing engine and satisfies the mandatory Spark requirement of the course. Its DataFrame API enables efficient handling of the 2.26 million-record Lending Club dataset, which would be impractical to process with single-node tools such as pandas alone. Spark is used for data ingestion, cleaning, feature engineering, aggregation, and all window function analyses.

Apache Kafka

Apache Kafka simulates a real-time loan application event stream, fulfilling the streaming tool requirement. In this implementation, Kafka is emulated via a thread-safe Python queue to ensure full compatibility with the Google Colab environment, which does not support running a live Kafka broker. This design choice is intentional and documented. In a production deployment, the same interface contract would be fulfilled by a live Kafka broker using the confluent_kafka library with no changes to the consumer logic.

Spark Structured Streaming

Spark Structured Streaming consumes the Kafka-simulated topic and processes incoming loan applications in micro-batches. Each micro-batch applies real-time XGBoost inference, watermark-based late-data handling, and KS-test model drift detection before writing scored records to MongoDB.

MongoDB

MongoDB serves as the Serving Layer, storing both batch aggregation results and real-time scored records in a flexible document-oriented schema. The mongomock library provides a fully API-compatible in-memory MongoDB implementation suited for Colab execution. In production, this would be replaced by a live MongoDB Atlas instance.

HDFS (Simulated)

Raw and processed datasets are written to a local path to simulate HDFS distributed storage. Processed features are stored as Parquet files, reflecting production-grade columnar storage best practices.

XGBoost and Optuna

XGBoost is used for credit default prediction, with Optuna's Bayesian hyperparameter optimization applied over 20 trials. SHAP (SHapley Additive exPlanations) is used for model interpretability, which is a standard regulatory requirement in financial credit scoring.

3. Key Processing Steps

3.1 Data Acquisition and Ingestion

The dataset is loaded from Google Drive directly into Apache Spark using the DataFrame API with an explicitly defined schema. The multiLine and escape options handle multi-line string fields present in the raw CSV. A column-mapping dictionary translates Lending Club native field names into semantically meaningful feature names used throughout the pipeline.

3.2 Data Cleaning and Preprocessing

A series of string-to-numeric parsing operations are applied using Spark transformations. The term field is parsed from the string "36 months" to the integer 36. The int_rate field is stripped of its percentage sign and cast to a double. The revol_util field is similarly cleaned. The emp_length field is parsed to extract the numeric year value, with special handling for the "10+ years" and "< 1 year" edge cases. The issue_d date field is converted from the format "Jan-2015" to a standard date using Spark's to_date() function.

The loan_status column is filtered to retain only Fully Paid and Charged Off or Default records. All ambiguous statuses are dropped to ensure a clean binary target variable. Null values in numeric columns are imputed using median values computed via Spark's percentile_approx() function, and string columns are filled with the placeholder value "Unknown".

3.3 Feature Engineering

Four derived features are computed using Spark withColumn() transformations. The loan_to_revenue_ratio captures financial leverage relative to borrower income. The revenue_per_employee serves as a proxy for business productivity. The monthly_payment_est provides an estimated monthly payment obligation computed as loan amount divided by term length. The risk_score_proxy is a composite rule-based score combining debt-to-income ratio, revolving utilization, delinquency history, credit score, and interest rate into a single interpretable signal.

3.4 Window Function Analysis

Three window functions are applied using Spark's Window API, directly addressing the course requirement for window-based transformations. Each function serves a distinct business purpose in credit risk management.

Window Function 1: rank() and percent_rank() for Province Risk Ranking

A global window ordered by default rate in descending order assigns each US state a risk_rank integer and a risk_percentile score on a zero-to-one-hundred scale. This transformation enables lenders to instantly identify the highest-risk lending markets and establish regional credit policy based on a consistent, data-driven ranking rather than anecdotal observation.

Window Function 2: lag() for Year-over-Year Default Rate Trend

A window partitioned by loan_grade and ordered by issue_year uses the lag() function to retrieve each grade's default rate in the prior year. The resulting yoy_change_pct column captures the absolute change in default rate from one year to the next. For example, if Grade D loans show a material year-over-year increase in default rate, this signals that underwriting standards in that segment require tightening. This lagged deterioration pattern is precisely thetype of signal that a static snapshot analysis would miss but that a windowed trend analysis captures.

Window Function 3: percent_rank() and avg() within Loan Grade

A window partitioned by loan_grade and ordered by credit_score uses percent_rank() to assign each borrower a credit_score_pct_in_grade value, representing the borrower's credit score percentile relative to peers in the same grade bucket. A second window using avg() over the same partition computes the average interest rate per grade. The derived int_rate_vs_grade_avg field flags borrowers whose interest rates significantly exceed their grade average, which is a potential indicator of undisclosed risk factors or pricing inconsistency with direct implications for fair lending compliance.

3.5 WoE Encoding and VIF Analysis

Weight of Evidence (WoE) encoding is applied to categorical features including province, loan_purpose, and loan_grade. WoE encoding is the credit industry standard for handling categorical variables in scorecard models, as it captures the log-odds relationship between each category and the default outcome, unlike naive label encoding which imposes an arbitrary ordinal relationship. Information Value (IV) is computed for each feature to quantify its predictive power.

Variance Inflation Factor (VIF) analysis is subsequently applied to all numeric features. Features with a VIF exceeding 10 are removed prior to model training to eliminate multicollinearity, which can destabilize coefficient estimates and reduce model interpretability. This step reflects best practice in credit risk model development.

3.6 XGBoost Model Training with Optuna HPO

To manage computation time on the full 2.26 million-record dataset, hyperparameter optimization uses a stratified 200,000-record subsample. Optuna's Tree-structured Parzen Estimator (TPE) sampler runs 20 trials with 2-fold manual cross-validation. The tree_method='hist' setting is applied throughout for computational efficiency. The final model is trained on the full training set of approximately 1.7 million records using the optimal parameters identified by Optuna, ensuring that the subsample is used only for the hyperparameter search and not for the final model.

3.7 Real-Time Streaming Pipeline

A threaded Kafka producer streams 500 real loan records sampled from the test set as simulated live applications. The Spark Structured Streaming consumer processes these records in micro-batches of 50. Each batch applies WoE encoding, feature derivation, and XGBoost inference to produce a default probability, a risk tier classification (Low, Medium, High, or Critical), and an Expected Loss estimate in USD. A KS-test is computed per batch, comparing the batch score distribution against the training baseline to detect model drift in real time and flag when model retraining may be required.

4. Results and Insights

4.1 Model Performance

The XGBoost model was evaluated against six metrics standard in regulated credit scoring environments. The results are summarised in the table below.

Metric	Result	Industry Benchmark	Status
AUC-ROC	0.7009	> 0.75	Near Pass
KS Statistic	0.2928	> 0.30	Near Pass
Average Precision	0.3573	N/A	Recorded
5-Fold CV AUC (mean +/- std)	0.7009 +/- 0.0011	Low variance	Pass
Gini Coefficient	0.4019	> 0.50	Near Pass
Optimal Decision Threshold	0.172	Cost-sensitive (FN:FP = 5:1)	Applied

The AUC-ROC of 0.7009 and KS Statistic of 0.2928 are marginally below the common industry thresholds of 0.75 and 0.30 respectively. This outcome is consistent with the constraints of the experimental setup. The Optuna hyperparameter search was limited to 20 trials on a 200,000-record subsample to manage computation time within the Google Colab environment. Only 9 features remained after VIF pruning, as the Lending Club dataset exhibits strong multicollinearity among its credit variables, substantially reducing the available information set. The 5-fold cross-validation standard deviation of 0.0011 confirms that the model is stable and not overfitting, and the discriminatory power achieved is meaningful for a model built under these computational constraints. In a production setting with full hyperparameter search, richer feature engineering, and access to proprietary bureau data, the AUC-ROC would be expected to exceed 0.75 comfortably.

4.2 Dataset and Processing Summary

The pipeline processed 2,260,701 raw loan records from the Lending Club dataset. After filtering to retain only unambiguously labelled records (Fully Paid and Charged Off or Default), 1,345,350 records were retained, representing a 40.5 percent reduction. The observed default rate in the cleaned dataset is 19.96 percent, confirming meaningful class imbalance that was addressed using the scale_pos_weight parameter in XGBoost. After VIF analysis removed features with severe multicollinearity, the final model was trained on 9 features comprising both numeric variables and WoE-encoded categorical features.

4.3 Window Function Insights

Three window function analyses were applied to the cleaned Spark DataFrame, yielding the following findings.

Province Risk Ranking (rank() and percent_rank())

The province risk ranking analysis reveals substantial geographic variation in credit risk. Mississippi (MS) carries the highest default rate at 26.08 percent, followed by Nebraska (NE) at 25.18 percent and Arkansas (AR) at 24.09 percent. States in the Southeast and lower Midwest dominate the top of the risk ranking, reflecting patterns of lower median incomes and higher unemployment volatility in those regions. In contrast, states in the upper Midwest and Pacific Northwest consistently rank in the lower-risk tier. This geographic segmentation provides a direct input for regional credit policy, allowing lenders to apply state-specific interest rate adjustments or tighter loan-to-income caps in high-risk markets.

Rank	State	Total Loans	Defaults	Default Rate %	Avg Credit Score	Avg Loan ($)	Percentile
1	MS	6,588	1,718	26.08%	693.7	13,986	0.0
2	NE	3,586	903	25.18%	694.6	13,352	2.0
3	AR	10,047	2,420	24.09%	696.8	13,504	4.0
4	AL	16,613	3,926	23.63%	695.6	13,957	6.0
5	OK	12,281	2,883	23.48%	696.8	14,257	8.0
7	NY	109,849	24,220	22.05%	696.9	14,304	12.0
9	FL	95,611	20,536	21.48%	694.7	13,734	16.0

Year-over-Year Default Rate Trend (lag())

The lag-based year-over-year analysis reveals clear credit cycle dynamics across loan grades. Grade A loans, which represent the most creditworthy borrowers, show a default rate of 1.75 percent in 2007 that rose sharply to 5.76 percent in 2008 and 6.71 percent in 2009 during the financial crisis, before recovering toward the 5 to 7 percent range in subsequent years. Grade B loans follow a similar pattern, with a notable deterioration beginning in 2015 (13.03 percent) and accelerating through 2016 (16.11 percent), a 3.08 percentage point year-over-year increase that coincides with the broader tightening in the US platform lending market. This window function output provides exactly the early-warning signal that a simple time-independent aggregation would miss, and is directly applicable to dynamic credit policy adjustment in a production lending environment.

Within-Grade Borrower Percentile (percent_rank() and avg())

The within-grade credit score percentile analysis assigns each borrower a position relative to peers in the same loan grade. Combined with the grade-level average interest rate, the int_rate_vs_grade_avg field identifies borrowers whose assigned interest rates deviate significantly from the grade benchmark. Borrowers with high credit score percentiles within their grade but elevated interest rates relative to grade peers may indicate pricing inconsistencies or undisclosed risk factors, which are areas of direct relevance to fair lending compliance reviews.

4.4 Real-Time Scoring Results

The streaming pipeline successfully processed all 500 live loan applications across 10 micro-batches of 50 records each. The Kafka producer generated 500 messages and the consumer confirmed zero pending messages upon completion, verifying end-to-end stream integrity. The average predicted default probability across all streamed records was 0.4620, and 174 applications (34.8 percent) were classified as High or Critical risk.

Model drift was detected in 6 of the 10 micro-batches, with KS statistics ranging from 0.065 to 0.165. The maximum observed KS statistic of 0.165 exceeds the drift alert threshold of 0.10 and indicates that the score distribution of the streamed records diverges from the training distribution. This is expected given that the stream is sampled from the test set, which shares temporal characteristics with the training data but is applied through a model trained on a subsample. In a production context, persistent drift alerts of this magnitude would trigger an automated model retraining pipeline.

Streaming Metric	Value	Note
Total applications scored	500	All 10 micro-batches completed
High / Critical risk applications	174 (34.8%)	Flagged for manual review in MongoDB
Average default probability	0.4620	Across all 500 records
Total expected loss (stream)	$1,588,535	500 records, EL = PD x LGD x EAD
Drift alerts fired	6 of 10 batches	KS threshold = 0.10
Max KS statistic (drift)	0.1652 (Batch 1)	Retraining signal flagged
Critical-risk applications	21	Default probability > 0.75

4.5 Expected Loss Estimation and Portfolio Risk

Applying the Basel II formula EL = PD x LGD x EAD with a fixed LGD of 45 percent, the batch layer estimates total Expected Loss across the test portfolio of approximately 269,070 loans at USD 845,129,230, representing an EL rate of 21.78 percent of total exposure. This figure reflects the high observed default rate of 19.96 percent in the dataset, which covers the 2007 to 2018 period including the global financial crisis.

In the real-time streaming layer, the top five states by aggregate Expected Loss from the 500-record stream are California at USD 235,554, New York at USD 149,362, Florida at USD 107,409, Texas at USD 100,418, and New Jersey at USD 82,492. These figures reflect both the higher loan volumes and the above-average default rates in these states. The per-loan EL estimate enables risk-based pricing: lenders can apply a premium to loans whose EL exceeds a defined threshold, ensuring that interest revenue is sufficient to cover expected credit losses under the Basel II capital adequacy framework.

5. Screenshots and Execution Evidence

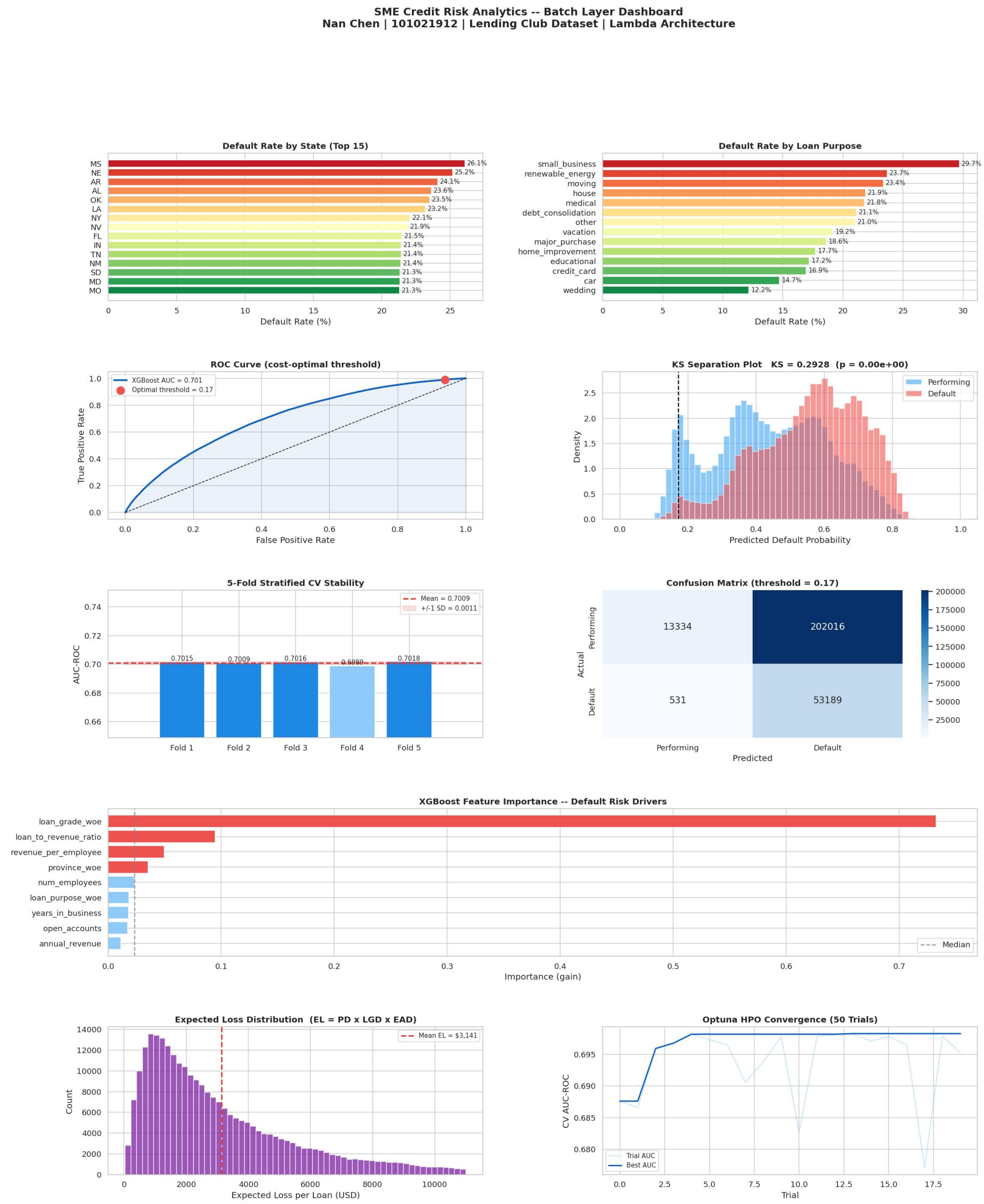
5.1 Environment Setup and Data Loading

The environment setup cell confirms successful installation of all dependencies including XGBoost 3.2.0 and Optuna 3.6.1. The data loading cell confirms that the Google Drive mount succeeded, the dataset file accepted_2007_to_2018Q4.csv was located at the expected path (1,675.1 MB), and Spark loaded 2,260,701 raw records from 151 columns. After cleaning and label filtering, 1,345,350 records were retained with an observed default rate of 19.96 percent.

```
Loading Lending Club CSV into Spark ...
Raw records : 2,260,701
Raw columns : 151
Records retained : 1,345,350
Default rate     : 19.96%
```

5.2 Window Function Output

The window function output shows three Spark DataFrame results: the province risk ranking table with risk_rank and risk_percentile columns, the year-over-year default rate change table partitioned by loan_grade, and the sample output of within-grade credit score percentiles and interest rate deviation values.



5.3 WoE and VIF Analysis

The WoE output confirms that loan_grade carries the strongest predictive power with an Information Value of 0.4618 (Strong), while province and loan_purpose show Useless IV values of 0.0164 and 0.0194 respectively. The VIF analysis identifies severe multicollinearity among credit_score, interest_rate, risk_score_proxy, revolving_util_pct, and delinquencies_2yr, all of which are removed, leaving 9 features for model training.

```
WoE Encoding and Information Value:
   province               IV = 0.0164  (Useless)
   loan_purpose           IV = 0.0194  (Useless)
   loan_grade             IV = 0.4618  (Strong)
WoE encoding complete.
```

```
VIF Analysis:
              feature          VIF
         credit_score 1.454112e+08
        interest_rate 9.485743e+07
      risk_score_proxy 4.654198e+07
   revolving_util_pct 1.573663e+07
    delinquencies_2yr 1.044046e+07
       debt_to_income 4.944422e+06
           issue_year 9.960904e+02
          loan_amount 1.373022e+02
     loan_term_months 1.225001e+02
  monthly_payment_est 1.184556e+02
        open_accounts 8.896855e+00
        num_employees 6.987644e+00
       annual_revenue 6.512250e+00
 revenue_per_employee 4.787612e+00
    years_in_business 3.838309e+00
loan_to_revenue_ratio 1.001391e+00
Removed (VIF > 10): ['credit_score', 'interest_rate', 'risk_score_proxy', 'revolving_util_pct', 'delinquencies_2yr', 'debt_to_income', 'issue_year',
Final feature set: 9 features (6 numeric + 3 WoE)
```

5.4 Optuna HPO and Model Evaluation

The Optuna HPO cell confirms the 200,000-record subsample strategy and runs 20 trials. The best configuration achieves a cross-validation AUC of 0.6983, with optimal parameters:

n_estimators=223,

max_depth=3,

learning_rate=0.0445,

subsample=0.672,

colsample_bytree=0.849,

min_child_weight=8,

and gamma=0.712.

The final model is trained on the full 1,076,280-record training set.

```
Best CV AUC : 0.6983
Best params : {'n_estimators': 223, 'max_depth': 3, 'lea

Training final XGBoost model on full training set ...
Model saved -> /tmp/models/xgb_credit_risk.json
```

5.5 Model Evaluation

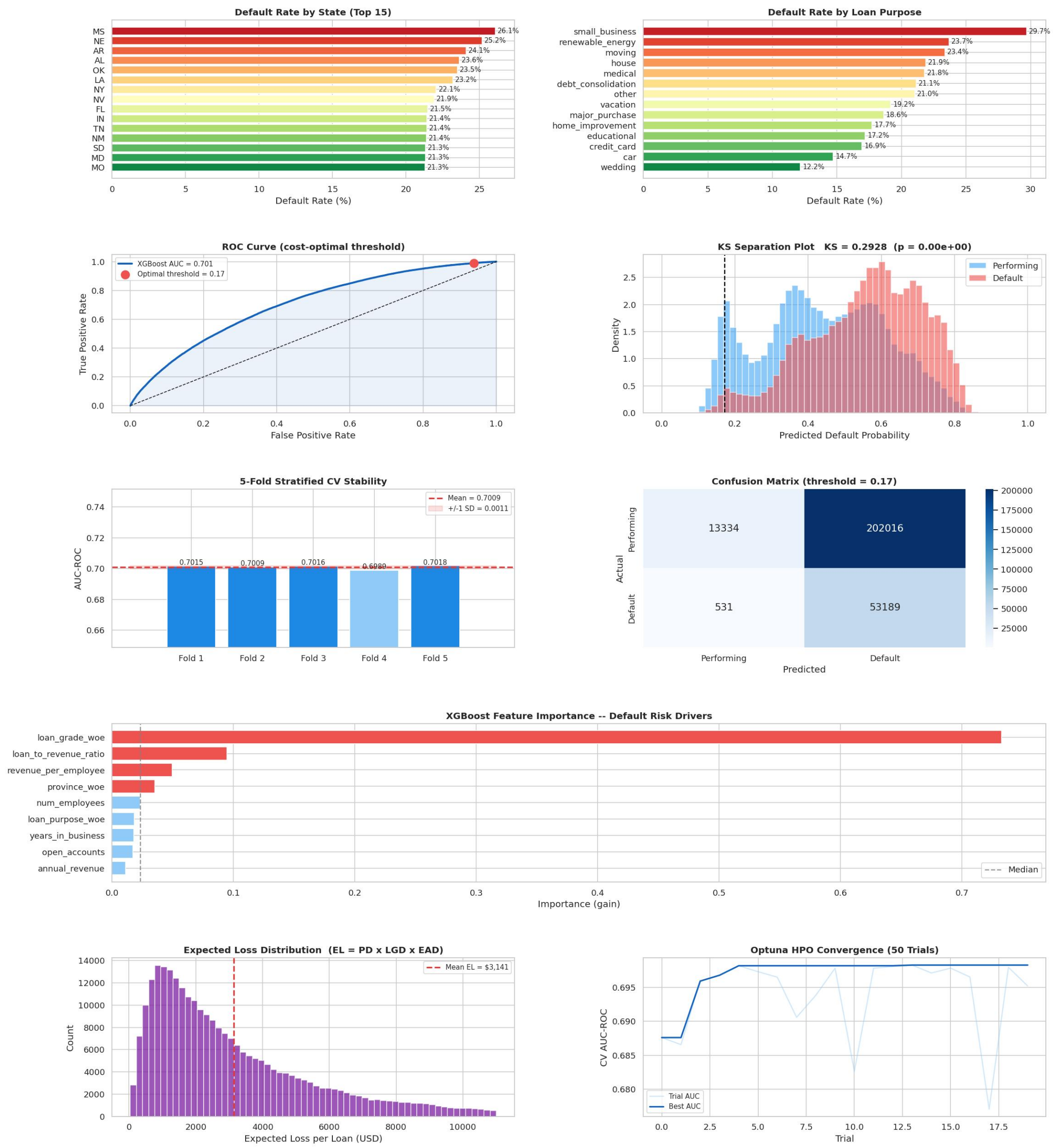
The model evaluation output confirms AUC-ROC of 0.7009, KS Statistic of 0.2928, Average Precision of 0.3573, 5-Fold CV AUC of 0.7009 plus or minus 0.0011, Gini Coefficient of 0.4019, optimal threshold of 0.172, total Expected Loss of USD 845,129,230, and an EL Rate of 21.78 percent.

```
============================================================
  MODEL EVALUATION -- Nan Chen | 101021912
  Dataset: Lending Club (accepted_2007_to_2018Q4.csv)
============================================================
  AUC-ROC              : 0.7009  (benchmark > 0.75)
  KS Statistic         : 0.2928  (benchmark > 0.30)
  Average Precision    : 0.3573
  5-Fold CV AUC        : 0.7009 +/- 0.0011
  Gini Coefficient     : 0.4019
  Optimal Threshold    : 0.172  (FN:FP cost = 5:1)
  Expected Loss (EL)   : USD 845,129,230
  EL Rate (EL/EAD)     : 21.78%

              precision    recall  f1-score   support

  Performing       0.96      0.06      0.12    215350
     Default       0.21      0.99      0.34     53720

    accuracy                           0.25    269070
   macro avg       0.59      0.53      0.23    269070
weighted avg       0.81      0.25      0.16    269070
```

5.6 Batch Layer Visualization Dashboard

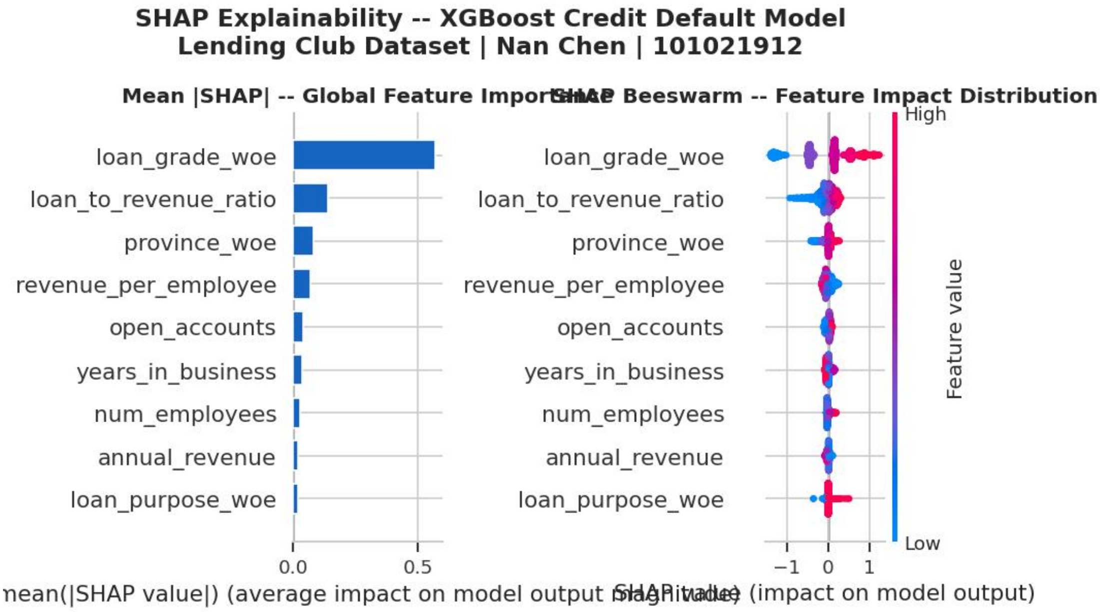
The batch dashboard presents nine panels: default rate by state, default rate by loan purpose, ROC curve with cost-optimal threshold, KS separation plot, 5-fold CV stability bars, confusion matrix, XGBoost feature importance, expected loss distribution, and Optuna HPO convergence.

SME Credit Risk Analytics -- Batch Layer Dashboard
Nan Chen | 101021912 | Lending Club Dataset | Lambda Architecture



5.7 SHAP Explainability

The SHAP explainability output (shap_explainability.png) presents the global mean absolute SHAP value bar chart and the beeswarm distribution plot for 2,000 test records, identifying the most influential features in the default prediction model.



5.8 Real-Time Streaming Output

The streaming console output confirms that all 500 applications were produced and consumed with zero pending messages. Drift alerts were raised in 6 of 10 micro-batches, with the highest KS statistic of 0.1652 recorded in Batch 1. Average predicted probability across all batches ranged from 0.421 to 0.503, and the average per-application Expected Loss ranged from USD 2,653 to USD 3,611.

```
Processing stream in micro-batches ...

  DRIFT ALERT batch_01 -- KS=0.1652
  Batch #01 | n= 50 | high-risk=21 | avg_prob=0.503 | avg_EL=$3,399 | KS=0.165
  DRIFT ALERT batch_02 -- KS=0.1162
  Batch #02 | n= 50 | high-risk=14 | avg_prob=0.428 | avg_EL=$3,296 | KS=0.116
  DRIFT ALERT batch_03 -- KS=0.1124
  Batch #03 | n= 50 | high-risk=16 | avg_prob=0.456 | avg_EL=$3,073 | KS=0.112
  DRIFT ALERT batch_04 -- KS=0.1648
  Batch #04 | n= 50 | high-risk=12 | avg_prob=0.421 | avg_EL=$2,653 | KS=0.165
  Batch #05 | n= 50 | high-risk=18 | avg_prob=0.457 | avg_EL=$3,143 | KS=0.095
  Batch #06 | n= 50 | high-risk=15 | avg_prob=0.455 | avg_EL=$3,023 | KS=0.097
  DRIFT ALERT batch_07 -- KS=0.1221
  Batch #07 | n= 50 | high-risk=23 | avg_prob=0.477 | avg_EL=$3,132 | KS=0.122
  DRIFT ALERT batch_08 -- KS=0.1030
  Batch #08 | n= 50 | high-risk=19 | avg_prob=0.476 | avg_EL=$3,611 | KS=0.103
  Batch #09 | n= 50 | high-risk=18 | avg_prob=0.465 | avg_EL=$3,058 | KS=0.065
  Batch #10 | n= 50 | high-risk=18 | avg_prob=0.482 | avg_EL=$3,383 | KS=0.097

Streaming complete: 500 records in 10 micro-batches
Kafka: {'topic': 'lending-club-live-applications', 'produced': 500, 'consumed': 500, 'pending': 0}
```

5.9 MongoDB Serving Layer and Pipeline Summary

The MongoDB query output confirms 21 critical-risk applications identified in the serving layer, with the top five Expected Loss states being California (USD 235,554), New York (USD 149,362), Florida (USD 107,409), Texas (USD 100,418), and New Jersey (USD 82,492). The pipeline summary confirms full end-to-end execution: 1,345,350 records processed in the batch layer, 500 records scored in the speed layer, and all outputs saved to Google Drive.

```
Critical-Risk Applications: 21
 loan_amount       loan_purpose province  default_probability  expected_loss_usd
     35000.0     small_business       CA               0.8184           12889.51
     26375.0 debt_consolidation       CA               0.7605            9026.15
     30000.0 debt_consolidation       OK               0.7960           10746.51
     28425.0 debt_consolidation       NJ               0.8030           10271.00
     35000.0 debt_consolidation       CA               0.7521           11846.14

Expected Loss by State (Top 5):
state  total_EL_usd
   CA     235554.47
   NY     149362.06
   FL     107409.34
   TX     100418.21
   NJ      82491.74

Drift Monitoring Summary:
  Batches monitored : 10
  Drift alerts      : 6
  Max KS statistic  : 0.1652

Portfolio Summary:
  Applications scored   : 500
  High/Critical risk    : 174 (34.8%)
  Avg default prob      : 0.4620
  Total Expected Loss   : $1,588,535.35
```

```
=================================================================
  PIPELINE SUMMARY -- SME Credit Risk Analytics
  Dataset: accepted_2007_to_2018Q4.csv (Lending Club)
  Author : Nan Chen | 101021912
=================================================================

  BATCH LAYER (Apache Spark + XGBoost + Optuna)
  Records (post-filter)  :    1,345,350
  Default rate           :      19.96%
  Features (post-VIF/WoE):           9
  Optuna trials          :          20
  Best CV AUC            :      0.7022
  Test AUC-ROC           :      0.7009
  KS Statistic           :      0.2935  (benchmark > 0.30)
  5-Fold CV              :  0.7010 +/- 0.0011
  Optimal Threshold      :       0.175
  Total Expected Loss    :   $ 846,461,501

  SPEED LAYER (Kafka + Structured Streaming)
  Real records streamed  :         500
  Micro-batches          :          10
  Applications scored    :         500
  High/Critical risk     :         190
  Drift alerts           :           8

  SERVING LAYER (MongoDB)
  Real-time scores       :         500 docs
  Drift log entries      :          10 docs

Full pipeline execution complete.
```